

Fuzzy Logic

Unit 2: Fuzzy Sets, Operations, Relations, Inference & Fuzzy Control

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Course Page:

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Outline

- 1 Fuzzy Sets and Membership Functions
- 2 Operations, Relations, Rules and Inference
- 3 Defuzzification, Controller Design & Applications
- 4 Summary and Practice

Learning Outcomes of Unit 2

After this unit, a student will be able to...

- 1 **Define** fuzzy sets and membership functions and contrast them with crisp sets.
- 2 **Apply** the standard operations (union, intersection, complement) and verify their properties.
- 3 **Construct** fuzzy relations and compose them (max–min composition).
- 4 **Formulate** fuzzy propositions, IF–THEN rules and implications.
- 5 **Perform** approximate reasoning using the compositional rule of inference.
- 6 **Defuzzify** a fuzzy output (centroid, MoM, etc.) and **design** a simple fuzzy logic controller.
- 7 **Relate** fuzzy logic to real-life societal applications.

Mapping to the Syllabus

A. Fuzzy logic, fuzzy sets, membership functions. **B.** Operations, relations, rules, propositions, implications, inferences. **C.** Defuzzification, controller design, societal applications.

Recap: Crisp Sets and the Characteristic Function

Definition

A **crisp** set $S \subseteq X$ has *binary* membership described by its **characteristic function**:

$$\mu_S(x) = \begin{cases} 1 & x \in S \\ 0 & x \notin S \end{cases}$$

- Every element either **belongs** or **does not**.
- Boundaries are **sharp** — no “in-between”.
- This is the world of *hard computing* (Unit 1).

Example — “Tall people” (crisp)

Let $tall = \{h \mid h \geq 180 \text{ cm}\}$.

Then 179.9 cm is **not** tall, 180.0 cm **is** tall. Absurd at the boundary!

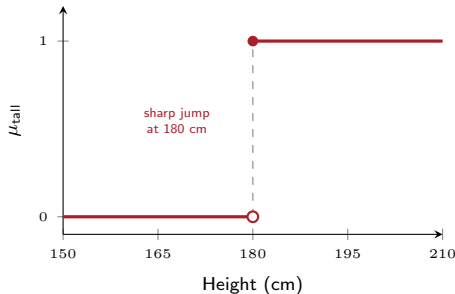


Figure: A crisp set has a discontinuous, all-or-nothing boundary.

Motivation: The World is Fuzzy

Crisp logic (two-valued)

Question: *Is the liquid colourless?*

Liquid	Answer
Water	Yes
Milk	No
Cola	No
Sprite	Yes

Only **True** / **False**.

Fuzzy logic (multi-valued)

Question: *Is the person honest?*

Person	Degree
Ankit	0.99 (extremely)
Rajesh	0.75 (very honest)
Santosh	0.55 (honest at times)
Kabita	0.35 (dishonest)

A **continuum** of truth $\in [0, 1]$.

Introduced by Lotfi A. Zadeh (1965) — “The closer you look at reality, the fuzzier it gets.”

What is a Fuzzy Set?

Formal Definition

A **fuzzy set** A in a universe of discourse X is a set of ordered pairs

$$A = \{(x, \mu_A(x)) \mid x \in X\}, \quad \mu_A : X \rightarrow [0, 1],$$

where $\mu_A(x)$ is the **grade (degree) of membership** of x in A .

- $\mu_A(x) = 1 \Rightarrow x$ is **fully** in A .
- $\mu_A(x) = 0 \Rightarrow x$ is **not** in A .
- $0 < \mu_A(x) < 1 \Rightarrow$ **partial** membership.

Notation

Discrete: $A = \sum_i \mu_A(x_i)/x_i$ (" $/$ " is a separator, not division).

Continuous: $A = \int_X \mu_A(x)/x$.

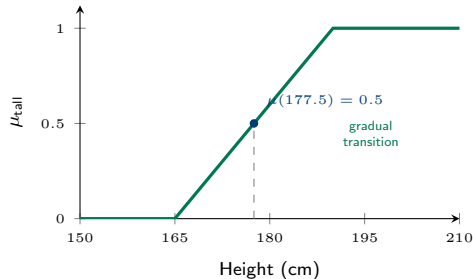


Figure: *Tall* as a fuzzy set — membership rises smoothly.

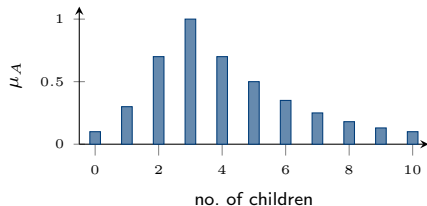
Key note

Fuzzy Set — Discrete and Continuous Universes

Discrete universe

$X = \{0, 1, \dots, 10\}$ (number of children),
 $A = \text{"Happy Family"}$.

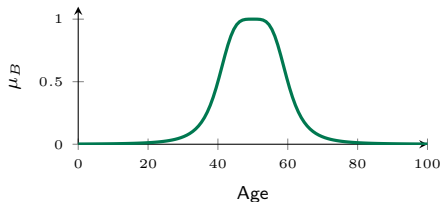
$$A = \frac{0.1}{0} + \frac{0.3}{1} + \frac{0.7}{2} + \frac{1.0}{3} + \frac{0.7}{4} + \dots + \frac{0.1}{10}$$



Continuous universe

$X = \mathbb{R}^+$ (age), $B = \text{"Middle-aged"}$.

$$\mu_B(x) = \frac{1}{1 + \left(\frac{x - 50}{10}\right)^4}$$



The membership function *is* the fuzzy set — choosing it well is the art of fuzzy modelling.

Key Terminologies of a Fuzzy Set

Definitions

- **Support:** $\text{supp}(A) = \{x \mid \mu_A(x) > 0\}$
- **Core:** $\text{core}(A) = \{x \mid \mu_A(x) = 1\}$
- **Height:** $h(A) = \max_x \mu_A(x)$
- **Normal:** A is normal if $h(A) = 1$; else *subnormal*
- **Crossover point:** $\{x \mid \mu_A(x) = 0.5\}$
- **α -cut:** $A_\alpha = \{x \mid \mu_A(x) \geq \alpha\}$ (strong: $> \alpha$)
- **Bandwidth:** distance between the two crossover points of a normal, convex set

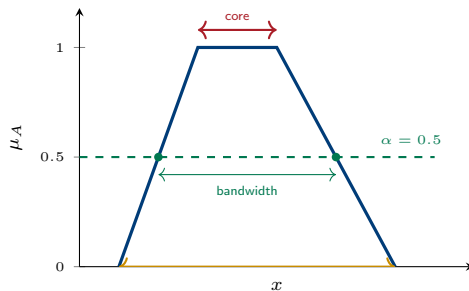
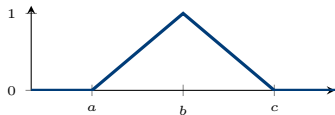


Figure: Support, core, α -cut, crossover points and bandwidth of a trapezoidal fuzzy set.

Standard Membership Functions (MFs)

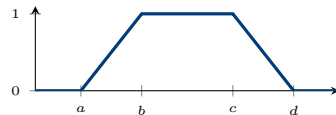
Triangular $\{a, b, c\}$

$$\mu(x) = \max\left(\min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right), 0\right)$$



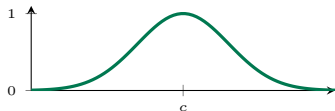
Trapezoidal $\{a, b, c, d\}$

$\mu = 1$ on $[b, c]$; linear ramps on $[a, b]$ and $[c, d]$; 0 outside.



Gaussian $\{c, \sigma\}$

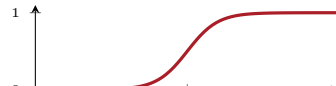
$$\mu(x) = \exp\left[-\frac{1}{2} \left(\frac{x-c}{\sigma}\right)^2\right]$$



Sigmoidal / Bell

Sigmoid: $\mu(x) = \frac{1}{1 + e^{-a(x-c)}}$

Gen. Bell: $\mu(x) = \frac{1}{1 + \left|\frac{x-c}{a}\right|^{2b}}$



Worked Example — Building a Triangular MF

“Warm” temperature = {15, 25, 35}

$$\mu_{\text{Warm}}(x) = \begin{cases} 0 & x \leq 15 \\ \frac{x - 15}{10} & 15 \leq x \leq 25 \\ \frac{35 - x}{10} & 25 \leq x \leq 35 \\ 0 & x \geq 35 \end{cases}$$

Evaluate

- $\mu_{\text{Warm}}(20) = \frac{20 - 15}{10} = \mathbf{0.5}$
- $\mu_{\text{Warm}}(28) = \frac{35 - 28}{10} = \mathbf{0.7}$
- $\mu_{\text{Warm}}(25) = \mathbf{1.0}$ (the core / peak)
- $\mu_{\text{Warm}}(40) = \mathbf{0}$

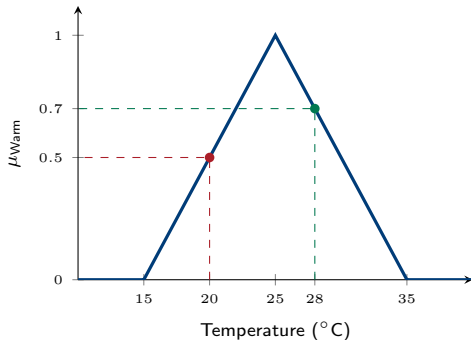


Figure: Reading membership grades straight off the triangle.

Quick Check — Round 1 (Fuzzy Sets & MFs)

[Q1]

Give the one-line difference between a crisp set and a fuzzy set.

[Q2]

For the triangular MF $Warm = \{15, 25, 35\}$, compute $\mu_{Warm}(30)$.

[Q3]

Define *support*, *core* and α -*cut* of a fuzzy set.

[Q4] True/False

“Every crisp set is a fuzzy set, but not every fuzzy set is crisp.”

Think for a minute — answers on the next slide.

Answers — Round 1

[A1]

A crisp set uses **binary** membership $\mu \in \{0, 1\}$ (belongs or not); a fuzzy set allows a **degree** of membership $\mu \in [0, 1]$.

[A2]

30 lies on the falling edge: $\mu_{\text{Warm}}(30) = \frac{35 - 30}{10} = \mathbf{0.5}$.

[A3]

Support = $\{x : \mu(x) > 0\}$ (all partially-belonging elements); *core* = $\{x : \mu(x) = 1\}$ (fully-belonging elements); α -*cut* $A_\alpha = \{x : \mu(x) \geq \alpha\}$ (a crisp set for a chosen threshold α).

[A4]

True. A crisp set is the special case where all membership values are exactly 0 or 1.

Standard Operations on Fuzzy Sets

Zadeh's operators

Complement: $\mu_{\bar{A}}(x) = 1 - \mu_A(x)$

Union: $\mu_{A \cup B}(x) = \max\{\mu_A(x), \mu_B(x)\}$

Intersect.: $\mu_{A \cap B}(x) = \min\{\mu_A(x), \mu_B(x)\}$

Others

- Bounded sum: $\min\{1, \mu_A + \mu_B\}$
- Bounded diff.: $\max\{0, \mu_A - \mu_B\}$
- Concentration A^2 ("very"): $[\mu_A]^2$
- Dilation $A^{0.5}$ ("more or less"): $[\mu_A]^{0.5}$

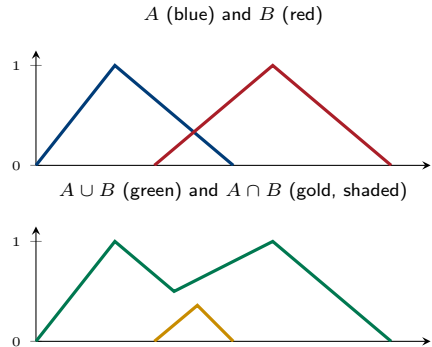


Figure: Union is the upper envelope; intersection the lower.

Worked Example — Operations on Discrete Fuzzy Sets

Given

On $X = \{1, 2, 3, 4, 5\}$: $A = \frac{0.2}{1} + \frac{0.5}{2} + \frac{0.8}{3} + \frac{1.0}{4} + \frac{0.4}{5}$, $B = \frac{0.6}{1} + \frac{0.9}{2} + \frac{0.3}{3} + \frac{0.2}{4} + \frac{0.7}{5}$.

x	1	2	3	4	5
μ_A	0.2	0.5	0.8	1.0	0.4
μ_B	0.6	0.9	0.3	0.2	0.7
$A \cup B = \max$	0.6	0.9	0.8	1.0	0.7
$A \cap B = \min$	0.2	0.5	0.3	0.2	0.4
$\bar{A} = 1 - \mu_A$	0.8	0.5	0.2	0.0	0.6
A^2 (very A)	0.04	0.25	0.64	1.00	0.16

Check — the excluded middle fails

At $x = 2$: $\min\{\mu_A, \mu_{\bar{A}}\} = \min\{0.5, 0.5\} = 0.5 \neq 0$. Hence $A \cap \bar{A} \neq \emptyset$ — **a fuzzy set overlaps its own complement.**

Properties of Fuzzy Set Operations

These still hold

- Commutativity, Associativity
- Distributivity
- Idempotence: $A \cup A = A$, $A \cap A = A$
- Involution: $\overline{(\bar{A})} = A$
- De Morgan's laws:
 $\overline{A \cup B} = \bar{A} \cap \bar{B}$,
 $\overline{A \cap B} = \bar{A} \cup \bar{B}$

These FAIL for fuzzy sets

- **Law of excluded middle:**
 $A \cup \bar{A} \neq X$ in general
- **Law of contradiction:**
 $A \cap \bar{A} \neq \emptyset$ in general

Why it matters

This single difference is what lets fuzzy logic represent *partial truth* — a statement can be “somewhat true and somewhat false” at once.

Verify De Morgan on the previous example at $x = 1$: $\overline{A \cup B}(1) = 1 - 0.6 = 0.4$;
 $\bar{A} \cap \bar{B}(1) = \min\{0.8, 0.4\} = 0.4$. ✓

Fuzzy Relations

Definition

A **fuzzy relation** R from X to Y is a fuzzy set on the product $X \times Y$:

$$R = \{((x, y), \mu_R(x, y)) \mid (x, y) \in X \times Y\}.$$

$\mu_R(x, y) \in [0, 1]$ measures the *strength of association* between x and y .

Relation matrix R

μ_R	y_1	y_2	y_3
x_1	0.8	0.1	0.0
x_2	0.3	1.0	0.6
x_3	0.0	0.4	0.9

Example — “ x is close to y ”

A relation on cities: $\mu_R(\text{Delhi}, \text{Noida}) = 0.9$,
 $\mu_R(\text{Delhi}, \text{Mumbai}) = 0.1$.

Cartesian product (rule “ A and B ”)

$$\mu_{A \times B}(x, y) = \min\{\mu_A(x), \mu_B(y)\}$$

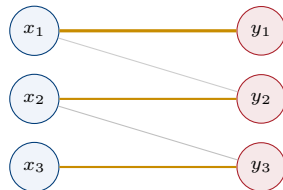


Figure: A fuzzy relation as a weighted bipartite graph.

Composition of Fuzzy Relations (Max–Min)

Definition

Given R on $X \times Y$ and S on $Y \times Z$, their **max–min composition** $R \circ S$ on $X \times Z$ is

$$\mu_{R \circ S}(x, z) = \max_{y \in Y} \min\{\mu_R(x, y), \mu_S(y, z)\}.$$

Data

$$R = \begin{pmatrix} 0.3 & 0.8 \\ 0.6 & 0.9 \end{pmatrix}, \quad S = \begin{pmatrix} 0.5 & 0.9 \\ 0.4 & 1.0 \end{pmatrix}$$

Compute entry (1, 1)

$$\begin{aligned} & \max\{\min(0.3, 0.5), \min(0.8, 0.4)\} \\ &= \max\{0.3, 0.4\} = \mathbf{0.4} \end{aligned}$$

Full result

$$(1, 1) = \max\{\min(0.3, 0.5), \min(0.8, 0.4)\} = 0.4$$

$$(1, 2) = \max\{\min(0.3, 0.9), \min(0.8, 1.0)\} = 0.8$$

$$(2, 1) = \max\{\min(0.6, 0.5), \min(0.9, 0.4)\} = 0.5$$

$$(2, 2) = \max\{\min(0.6, 0.9), \min(0.9, 1.0)\} = 0.9$$

$$R \circ S = \begin{pmatrix} 0.4 & 0.8 \\ 0.5 & 0.9 \end{pmatrix}$$

Composition is the engine of fuzzy inference (next).

Linguistic Variables, Propositions & Hedges

Linguistic variable

A variable whose **values are words**, not numbers.

$$T(\text{age}) = \{\text{young, old, very young, not old, ...}\}$$

Each term is a fuzzy set over the base variable (here, years).

Fuzzy proposition

“*x* is *A*” has a truth value $= \mu_A(x) \in [0, 1]$.

Connectives: AND = min, OR = max, NOT = $1 - \mu$.

Linguistic hedges (modifiers)

very A = μ_A^2 (concentration)

more or less A = $\mu_A^{0.5}$ (dilation)

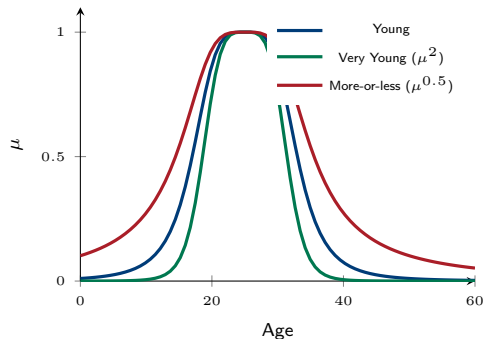


Figure: Hedges reshape the same base fuzzy set — “very” sharpens, “more or less” broadens.

Fuzzy IF–THEN Rules and Implication

Rule form

A **fuzzy rule** is a conditional statement

IF x is A **THEN** y is B ,

read as a fuzzy relation $R = A \rightarrow B$ on $X \times Y$.

Two common implications

- **Mamdani (min):**

$$\mu_R(x, y) = \min\{\mu_A(x), \mu_B(y)\}$$

- **Larsen (product):**

$$\mu_R(x, y) = \mu_A(x) \cdot \mu_B(y)$$

These *coupling* forms dominate in fuzzy control.

Example rule

IF temperature is **Cold** *THEN* heater is **High**.

- Antecedent fuzzy set: *Cold*
- Consequent fuzzy set: *High*
- A rule **base** is a set of such rules covering the input space.

Note

In logic, “ $A \rightarrow B$ ” can also use S -implications (e.g.

Approximate Reasoning — Generalised Modus Ponens

Classical vs. fuzzy

Classical modus ponens:

Premise: IF x is A THEN y is B ; x is A

Conclude: y is B .

Generalised (fuzzy):

Given a fact “ x is A' ” (close to A), infer

$$B' = A' \circ R, \quad \mu_{B'}(y) = \max_x \min\{\mu_{A'}(x), \mu_R(x, y)\}.$$

This is the compositional rule of inference (CRI)

The observed fuzzy input is *composed* with the rule relation to yield the fuzzy output.

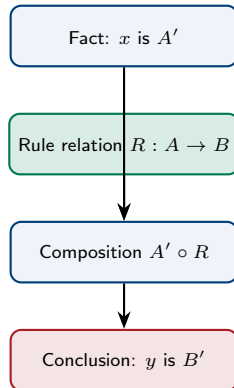


Figure: The compositional rule of inference.

If $A' = A$ exactly, fuzzy CRI reduces to the classical result $B' = B$.

Quick Check — Round 2 (Operations, Relations, Inference)

[Q1]

$\mu_A(x) = 0.7$, $\mu_B(x) = 0.4$. Find $\mu_{A \cup B}$, $\mu_{A \cap B}$ and $\mu_{\bar{A}}$ at x .

[Q2]

Which two classical laws *fail* for fuzzy sets? Give the reason.

[Q3]

Write the formula for max–min composition $\mu_{R \circ S}(x, z)$.

[Q4]

Express the hedge “very” as an operation on a membership function, and apply it to $\mu = 0.6$.

[A1]

$$\mu_{A \cup B} = \max\{0.7, 0.4\} = \mathbf{0.7}; \quad \mu_{A \cap B} = \min\{0.7, 0.4\} = \mathbf{0.4}; \quad \mu_{\bar{A}} = 1 - 0.7 = \mathbf{0.3}.$$

[A2]

The **law of excluded middle** ($A \cup \bar{A} = X$) and the **law of contradiction** ($A \cap \bar{A} = \emptyset$). Because μ can be strictly between 0 and 1, $\min\{\mu, 1 - \mu\}$ can be nonzero — a set overlaps its complement.

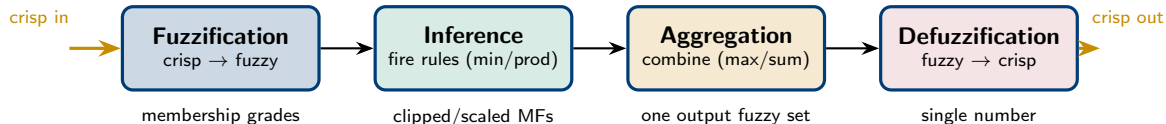
[A3]

$$\mu_{R \circ S}(x, z) = \max_y \min\{\mu_R(x, y), \mu_S(y, z)\}.$$

[A4]

very $A = \mu_A^2$ (concentration). At $\mu = 0.6$: $0.6^2 = \mathbf{0.36}$ — membership drops, as “very” is more demanding.

The Fuzzy Inference Pipeline



Two dominant fuzzy models

Mamdani: consequents are *fuzzy sets* $\Rightarrow$ needs defuzzification (intuitive, used in control). **Sugeno (TSK)**: consequents are *crisp functions* of the inputs $\Rightarrow$ output is a weighted average (efficient; detailed in Unit 3).

Why Defuzzify? & The Main Techniques

The need

Inference produces a *fuzzy* output set, but an actuator needs **one crisp number** (e.g. “fan speed = 42 rpm”).

Defuzzification maps the aggregated fuzzy set to a single value.

Common methods

- **Centroid / Centre of Gravity (CoG)** — balance point of the area
- **Bisector** — vertical line splitting area in two equal halves
- **Mean of Maximum (MoM)** — average of the peak-membership points
- **Smallest / Largest of Maximum** (SoM / LoM)
- **Weighted average** (Sugeno)

Formulas

Continuous centroid:

$$y^* = \frac{\int y \mu(y) dy}{\int \mu(y) dy}$$

Discrete centroid:

$$y^* = \frac{\sum_i y_i \mu(y_i)}{\sum_i \mu(y_i)}$$

Rule of thumb

CoG is *smooth* and most popular; MoM is *faster* but coarser. Choose by application, not by fashion.

Worked Example — Centroid Defuzzification

Discrete output fuzzy set

Fan-speed universe (rpm) with aggregated grades:

y_i	20	40	60	80	100
$\mu(y_i)$	0.2	0.6	0.9	0.4	0.1

Apply the centroid

$$\begin{aligned}\sum y_i \mu_i &= 20(0.2) + 40(0.6) + 60(0.9) \\ &\quad + 80(0.4) + 100(0.1) \\ &= 4 + 24 + 54 + 32 + 10 = 124 \\ \sum \mu_i &= 0.2 + 0.6 + 0.9 + 0.4 + 0.1 = 2.2\end{aligned}$$

$$y^* = \frac{124}{2.2} \approx \mathbf{56.4 \text{ rpm}}$$

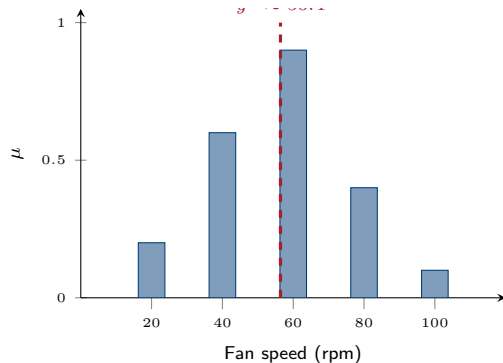
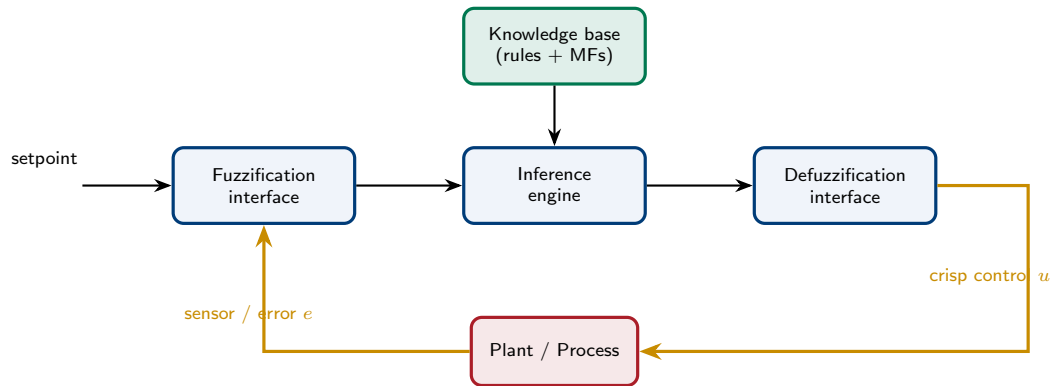


Figure: The centroid is the “balance point” of the output distribution.

Anatomy of a Fuzzy Logic Controller (FLC)



Design steps

(1) Identify input/output variables; (2) define linguistic terms & membership functions; (3) build the rule base from expert knowledge; (4) choose inference (Mamdani/Sugeno); (5) choose a defuzzifier; (6) **tune** MFs and rules by testing.

Design Walk-Through — A Fuzzy Air-Conditioner

Variables & terms

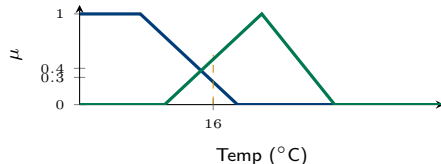
Input *Temperature*: {Cold, Cool, Pleasant, Warm, Hot}
Output *Fan speed*: {Stop, Slow, Medium, Fast, Blast}

Rule base (excerpt)

R1: IF Temp is **Cold** THEN Fan is **Stop**
R2: IF Temp is **Cool** THEN Fan is **Slow**
R3: IF Temp is **Pleasant** THEN Fan is **Medium**
R4: IF Temp is **Warm** THEN Fan is **Fast**
R5: IF Temp is **Hot** THEN Fan is **Blast**

Trace for $T = 16^{\circ}\text{C}$

- 1 **Fuzzify**: $\mu_{\text{Cool}}(16) = 0.3$, $\mu_{\text{Pleasant}}(16) = 0.4$
- 2 **Infer**: R2 fires at 0.3 $\rightarrow$ Slow; R3 fires at 0.4 $\rightarrow$ Medium
- 3 **Aggregate**: clip Slow at 0.3, Medium at 0.4; take max
- 4 **Defuzzify (CoG)** $\Rightarrow$ a crisp fan speed between Slow and Medium



Two-Input FLC — The Rule Matrix (FAM)

Fuzzy Associative Memory

A 2-input controller (e.g. *error* e and *change of error* Δe) is compactly written as a rule **matrix**. Each cell is the consequent term for that input combination.

Reading a cell

IF e is **NB** AND Δe is **ZE** THEN u is **NB**.
(NB = negative big, ZE = zero, PB = positive big.)

This is exactly how a fuzzy PID-like controller is specified — no plant model needed.

		$\Delta e \rightarrow$				
$e \downarrow$		NB	NS	ZE	PS	PB
NB		NB	NB	NM	NS	ZE
NS		NB	NM	NS	ZE	PS
ZE		NM	NS	ZE	PS	PM
PS		NS	ZE	PS	PM	PB
PB		ZE	PS	PM	PB	PB

Figure: A 5×5 FAM table — 25 rules on one page.
Diagonal symmetry $\Rightarrow$ balanced control action.

Fuzzy vs. Conventional (PID) Control

Conventional PID

$$u(t) = K_P e(t) + K_I \int_0^t e \, d\tau + K_D \frac{de}{dt}$$

- Needs a **precise plant model**.
- Struggles with non-linear / ill-modelled plants.

Fuzzy controller

- Encodes **expert linguistic rules**; *no explicit model*.
- Naturally handles non-linearity and multiple inputs.
- Mamdani (linguistic) or Sugeno (computationally light).

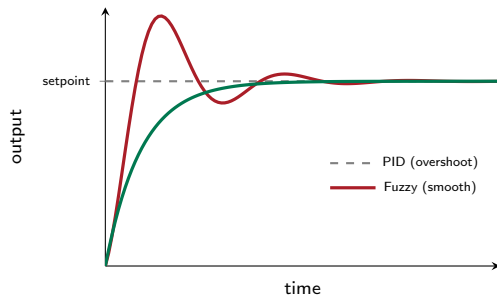


Figure: A well-tuned fuzzy controller can reach the setpoint with less overshoot on a non-linear plant.

Fuzzy Logic vs. Probability — Don't Confuse Them

Aspect	Fuzzy Logic	Probability
Models	Vagueness / imprecision	Randomness / uncertainty
Statement	"The patient has a <i>mild</i> fever"	"60% chance India wins"
Nature of category	The <i>category</i> itself is vague	The outcome is crisp; the <i>chance</i> varies
Value means	Degree of <i>belonging</i>	Frequency / likelihood of an <i>event</i>
Sums to 1?	No — memberships need not sum to 1	Yes — probabilities of a partition sum to 1
Based on	Expert judgement / perception	Recorded data / axioms

One-line memory hook

Fuzzy answers "**how much** does it belong?"; Probability answers "**how likely** is it to occur?" — they are complementary, not rival.

Societal Applications of Fuzzy Logic (1)

Home & consumer

- Washing machines: load/dirt sensing → water, time
- Air conditioners & refrigerators: smooth, energy-saving control
- Cameras: fuzzy auto-focus and exposure
- Rice cookers, microwave ovens

Transport & infrastructure

- **Sendai Subway (1987)** — fuzzy automatic train operation
- Fuzzy ABS, automatic transmission
- **Traffic-signal control** in congested cities
- Elevator (lift) group scheduling

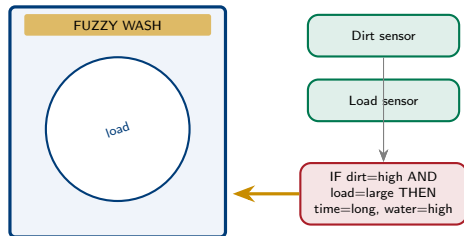


Figure: Sensor → fuzzy rules → actuator, the pattern behind most consumer fuzzy products.

Societal Applications of Fuzzy Logic (2)

Health & public services

- **Medical diagnosis** from vague symptoms (“mild”, “severe”) — remains *explainable* to clinicians
- Drug-dosage & anaesthesia-depth control
- Disaster warning: flood & fire risk indices
- Water-treatment and irrigation control

Why fuzzy suits society-scale problems

- Human judgements are **linguistic**, not numeric.
- Data is **noisy, partial and uncertain**.
- Decisions must be **transparent** and defensible.
- “Good enough, robust and cheap” beats “exact but impossible”.

Industry & environment

- Cement kilns, steel furnaces, chemical processes
- Power-system load & voltage regulation
- Air-quality indexing, weather nowcasting
- Fuzzy decision support for policy & resource allocation

Local design prompt

A fuzzy **traffic controller** for a Greater Noida junction reads *queue length* and *waiting time* to extend the green — adapting to real traffic a fixed timer cannot.

Advantages & Limitations of Fuzzy Logic

Advantages

- Captures **human/expert knowledge** directly as rules.
- **No precise model** of the plant required.
- Handles **non-linear, imprecise** systems gracefully.
- **Transparent & interpretable** (unlike a neural black box).
- Cheap sensors, low computational cost.
- Robust — **graceful degradation**.

Limitations

- Rules & membership functions need **expert tuning** (trial-and-error).
- No built-in **learning** — MFs don't self-adapt (motivates ANFIS, Unit 3).
- No formal **stability proof** in general.
- Rule count **explodes** with many inputs (curse of dimensionality).
- Choice of operators/defuzzifier is **application-specific**.

Bridge to Unit 3: couple fuzzy rules with neural learning $\Rightarrow$ *ANFIS* — rules that tune themselves.

Quick Check — Round 3 (Defuzzification & Control)

[Q1]

Why is defuzzification necessary in a Mamdani controller?

[Q2]

For outputs $\mu(20) = 0.4$, $\mu(40) = 0.8$, $\mu(60) = 0.2$, compute the centroid.

[Q3]

Name the four functional blocks of a fuzzy logic controller in order.

[Q4] MCQ

Which is *not* a defuzzification method?

(a) Centroid (b) Mean of Maximum (c) Bisector (d) Backpropagation

[A1]

Inference yields a *fuzzy* output set, but the actuator needs a single **crisp** command; defuzzification collapses the fuzzy set to one number.

[A2]

$$y^* = \frac{20(0.4) + 40(0.8) + 60(0.2)}{0.4 + 0.8 + 0.2} = \frac{8 + 32 + 12}{1.4} = \frac{52}{1.4} \approx \mathbf{37.1}.$$

[A3]

Fuzzification → **Inference engine** (with the *knowledge base*) → **Aggregation** → **Defuzzification**.

[A4]

(d) **Backpropagation** — that is a neural-network training rule, not a defuzzification method.

Summary of Unit 2

What we covered

- 1 **Fuzzy sets** — graded membership $\mu(x) \in [0, 1]$ generalising crisp sets.
- 2 **Membership functions** — triangular, trapezoidal, Gaussian, bell, sigmoid; support, core, α -cut, bandwidth.
- 3 **Operations** — union (max), intersection (min), complement; De Morgan holds, excluded middle fails.
- 4 **Relations** — fuzzy relations and max-min composition.
- 5 **Rules & inference** — linguistic variables, hedges, IF-THEN implication, generalised modus ponens (CRI).
- 6 **Defuzzification** — centroid, MoM, bisector; the fuzzy inference pipeline.
- 7 **Fuzzy control** — FLC anatomy, FAM rule matrix, fuzzy vs. PID.
- 8 **Applications** — appliances, transport, medicine, industry, public services.

Fuzzy logic lets machines reason with words the way humans do.

Practice Questions (Exam Oriented)

Short answer (2–5 marks)

- 1 Define a fuzzy set and its membership function. How does it differ from a crisp set?
- 2 State and illustrate the standard fuzzy operations (union, intersection, complement).
- 3 Which classical laws fail for fuzzy sets, and why?
- 4 Define support, core, height, normality and α -cut.
- 5 Distinguish fuzzy logic from probability with one example each.

Long answer (8–10 marks)

- 6 Given two discrete fuzzy sets, compute union, intersection, complement and “very”.
- 7 Explain max–min composition and solve a 2×2 numerical example.
- 8 Describe the fuzzy inference pipeline and defuzzify a given output by the centroid.
- 9 Draw and explain the block diagram of a fuzzy logic controller; design a rule base for a fuzzy air-conditioner.
- 10 Discuss four societal applications of fuzzy logic, stating why fuzzy suits each.

Think & Explore (Take-Home)

Mini design task

Design (on paper) a **fuzzy fan/heater controller**:

- Input: *room temperature* (5 terms)
- Output: *heater power* (5 terms)
- Sketch the membership functions
- Write 5 rules and defuzzify one case by the centroid

Coding (self-learning)

Implement the same controller in Python using `scikit-fuzzy` (`skfuzzy`) and compare with your hand calculation.

Reading & reflection

- Klir & Yuan, chapters on fuzzy sets, operations and relations.
- Ross, *Fuzzy Logic with Engineering Applications* — controller design.
- Prepare for **Unit 3**: Mamdani, Sugeno, Tsukamoto models and ANFIS.

Reminder

Assignment 1 covers Units 1 & 2 — handwritten, step-by-step, submitted as a single PDF via the course page.

References

Textbooks (as per the course page)

- ① G. J. Klir and B. Yuan, *Fuzzy Sets and Fuzzy Logic: Theory and Applications*, Prentice Hall. **[Main text]**
- ② T. J. Ross, *Fuzzy Logic with Engineering Applications*, McGraw Hill.
- ③ J.-S. R. Jang, C.-T. Sun and E. Mizutani, *Neuro-Fuzzy and Soft Computing*, Prentice Hall.

Seminal papers

- ④ L. A. Zadeh, "Fuzzy Sets," *Information and Control*, 8(3):338–353, 1965.
- ⑤ L. A. Zadeh, "Outline of a New Approach to the Analysis of Complex Systems and Decision Processes," *IEEE Trans. SMC*, 1973.
- ⑥ E. H. Mamdani and S. Assilian, "An experiment in linguistic synthesis with a fuzzy logic controller," *Int. J. Man-Machine Studies*, 1975.

NPTEL: *Soft Computing*, IIT Kharagpur · Course page:

<https://www.gcjana.in/courses/shardauniversity/2601/soft-computing/>

Thank You

Any Questions?

*“As complexity rises, precise statements lose meaning
and meaningful statements lose precision.”*

— Lotfi A. Zadeh (the Principle of Incompatibility)

Post your doubts on the course page → gcjana.in/.../soft-computing/#Post-doubts